10-Aug-2026

Scheduling of Projects – CPM

CE716 : Project Management and Controls

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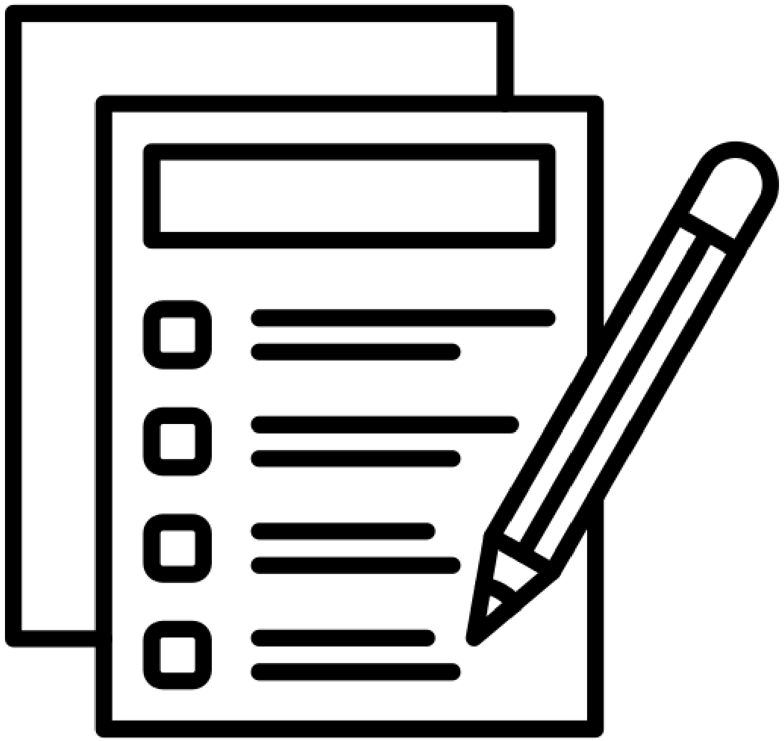
Department of Civil Engineering

Agenda

- Discussion on incorporating Safety
- Need for Network Diagram
- Types of Network Diagram
- Critical Path Method (CPM)
- Practical application of CPM



Practical Application - (Discussion)

How to make Tea (Black Tea)?

Activity	0	1	2	3	4	5	6	7	8	9	10
Find saucepan											
Boil water											
Find Tea leaves											
Add tea – let it simmer											
Filter and pour											

If there is so much safety incorporated in the Schedule – Why do projects get delayed?

- People who finish early won't report it
- Even if they report – the person supposed to do the next step will believe there is sufficient time.
- The student's syndrome (If there is more time- you tend to start late)
- Dependency in project – delays accumulate but advances get lost
- Impact of multi-tasking – its effect on lead time

Incorporating Safety in Schedules

Explore how to create Gantt Chart in Excel

Network Diagram

Need for Network Diagram

Why so much fuss? – Let’s try !

Can you estimate the total duration for this very small project?

You are tasked to construct ten identical columns of height 3m each. The site clearance and surveying will be undertaken in two phases (Phase 1 & 2), each taking around 2 days. The concreting of the columns in Phase 2 can only start after completion in Phase 1. However, other activities in Phase 2, like surveying, excavation, foundation shuttering, foundation reinforcement, and foundation concreting, can be carried out after the foundations in Phase 1 are complete. The excavation, shuttering, reinforcement, and concreting (up to 1.5 m height) for each column will take 3 days, 4 days, 2 days, and 1 day, respectively. Formwork, reinforcement, and concrete for each column foundation will take 1, 1, and 2 days, respectively. Each column pour is allowed to cure for 2 days before the next pour. Phase 1 consists of 4 columns and Phase 2 consists of 6 columns.

Why so much fuss? – Let's try !

Can you estimate the total duration for this very small project? (Assuming no restriction of labor, plant, or material (LMP).

How much manpower (skilled and unskilled workers) is needed for this project?

How many days will it take to complete the project if we decrease the maximum manpower required by 10%?

So on……….

Event, Activity, Dummy Activity

- Event
  - Point in time ;
  - Say start or end of an activity

- Activity –
  - consumes finite resources to achieve specific result.
  - Occurs between two events; has specific duration

- Dummy activity
  - Exemption to above rules; Has zero duration
  - Consumes zero resources and no tangible output

Precedence

- Logical relationship

- Implies an activity needs one or more activity to be completed before it can start

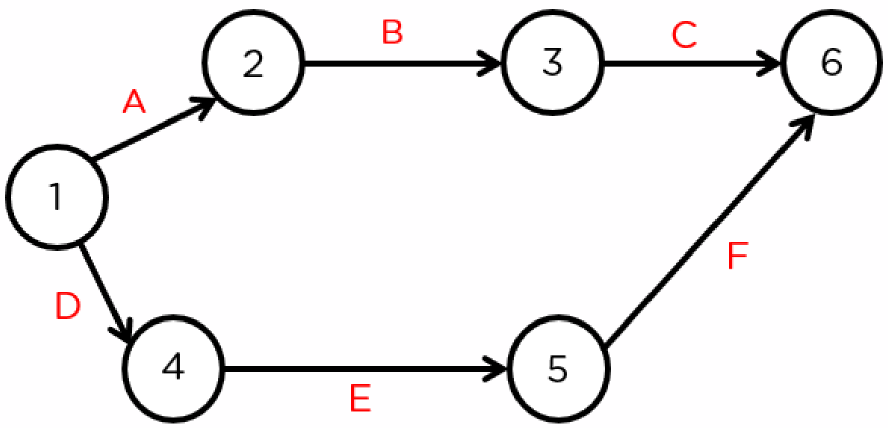
- Four common types of precedence relationships
  - Finish to start (FS)
  - Start to start (SS)
  - Finish to finish (FF)
  - Start to finish (SF)

Network diagram

- Shows logical dependence between activities

- Uses nodes and arrows to represent activities

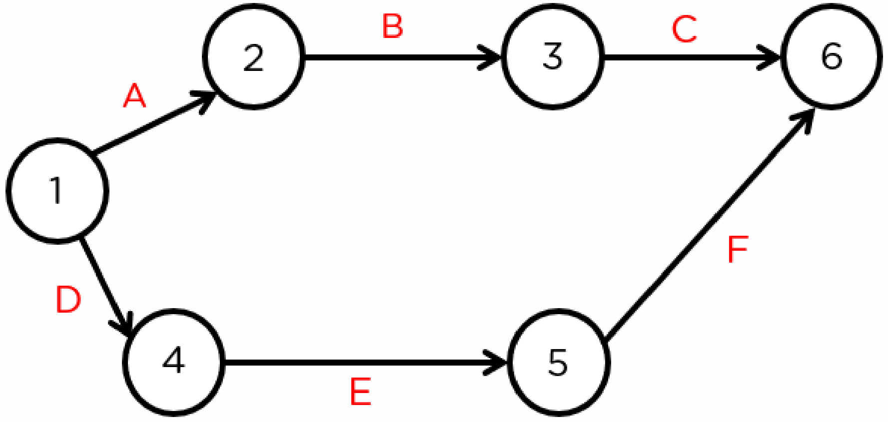
- Graphical representation of the project



Network diagram – AOA vs. AON

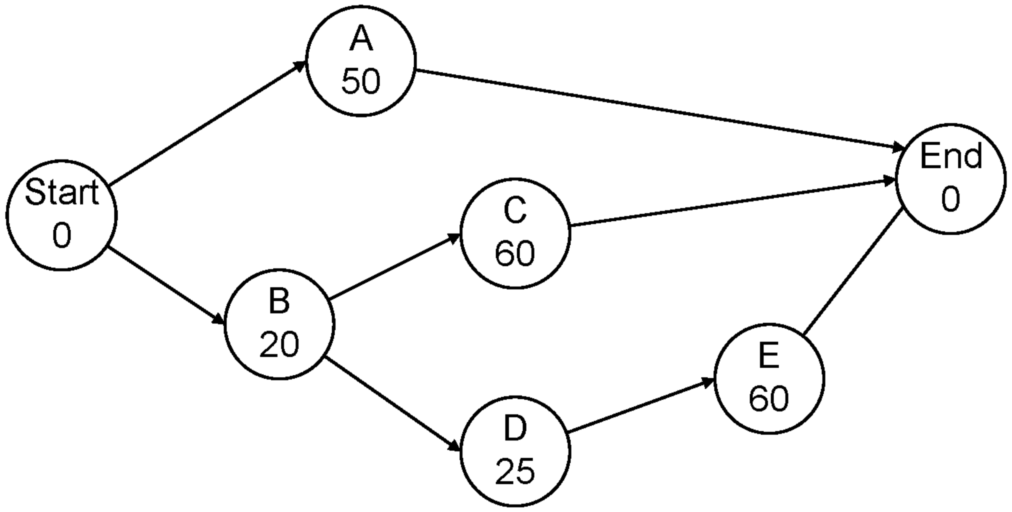
AOA – Activity on arrow

- Activity defined using unique pair of nodes – Example A_{12} or B_{23}



AON – Activity on node

- Duration of activity written within the Node



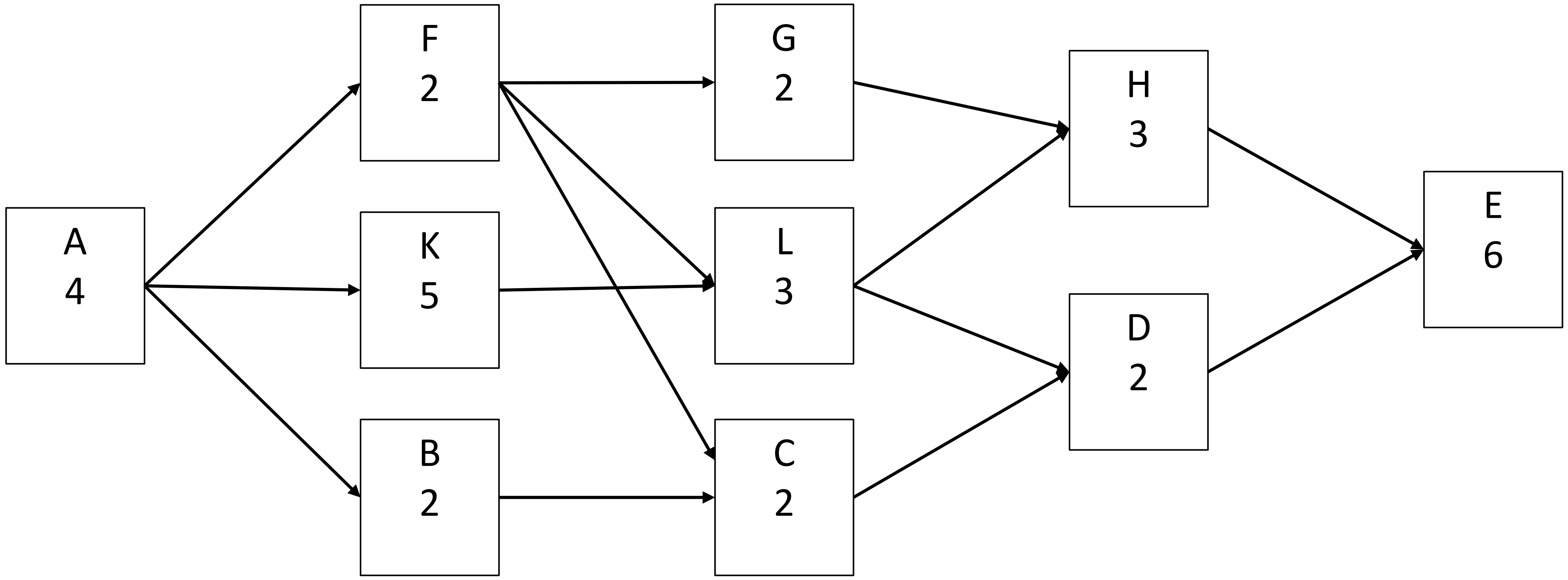
Example 1 – Create a Network Diagram

Activity	Duration (in weeks)	Preceding activity (FS)
A	4	-
B	2	A
C	2	B
D	7	A
E	8	C,D
F	2	E

Example 2 – Create a Network Diagram

Activity	Duration (in weeks)	Preceding activity (FS)
A	4	-
B	2	A
C	2	B, F
D	2	C, L
E	6	D, H
F	2	A
G	2	F
H	3	G, L
K	5	A
L	3	F, K

Example 2 - Solution



Critical Path Method

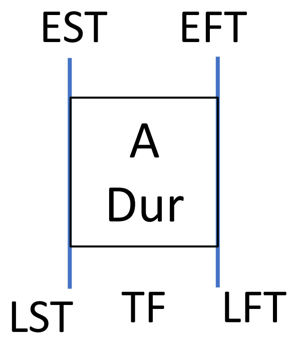
Few key terms & shortforms

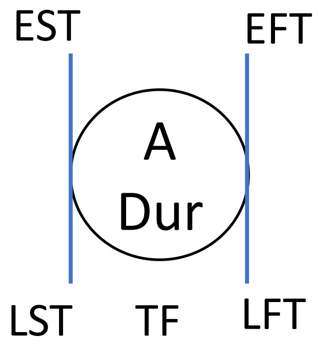
- D or Dur – Estimated duration of an activity
- EST – Earliest start time of an activity
- EFT – Earliest finish time of an activity

  EFT = EST + Dur

- LFT – Latest finish time of an activity (latest time an activity must complete so that there is no delay in the project)
- LST – Latest start time (latest time an activity must start to avoid delay)

  LST = LFT - Dur





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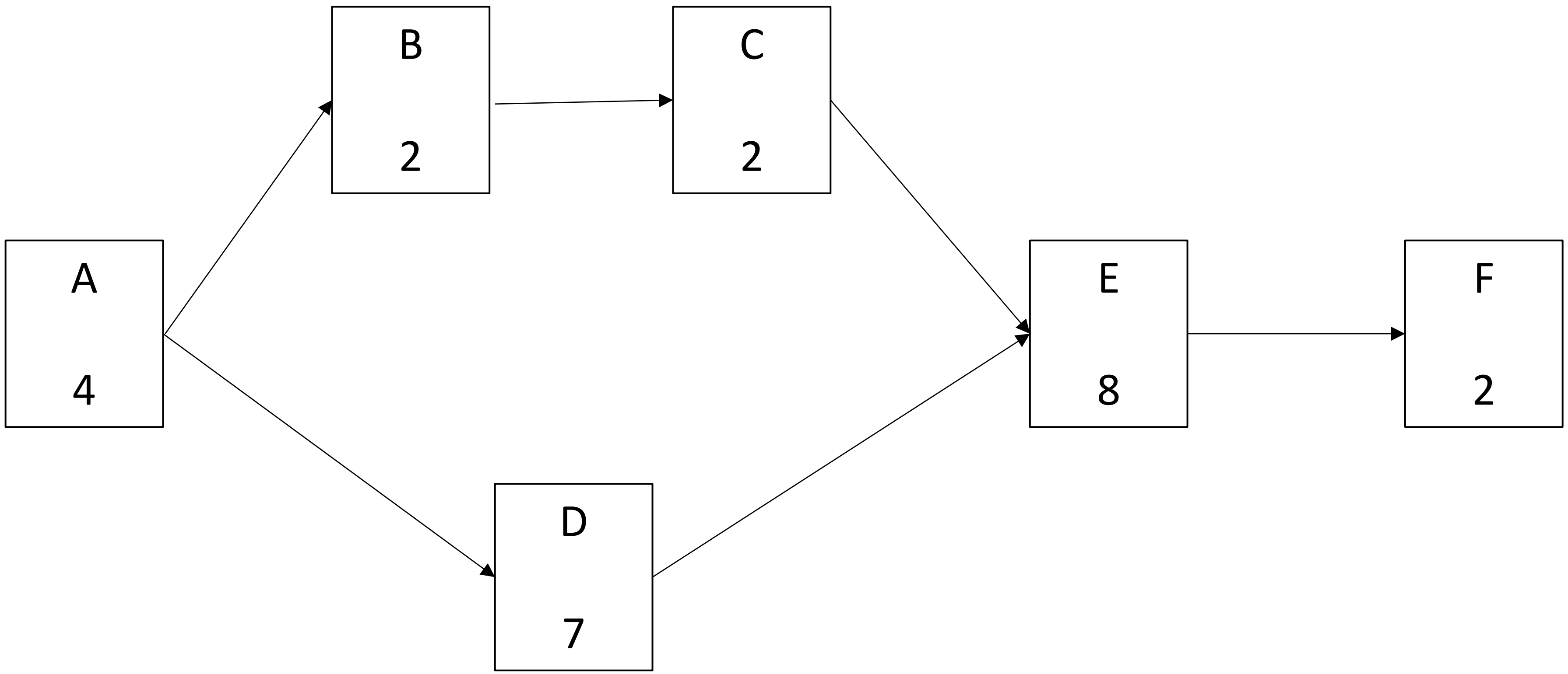
Forward pass

- Forward pass moves from ‘start’ node towards the ‘finish’ node
- Used to calculate earliest occurrence times of all the activities
- Typically, we consider project starts at t=0.
- The earliest start time for any node is the **Maximum** time taken to reach that node from all the activities (arrows) before it

Exercise 3 – Carry out forward pass on this network



Backward pass

- Backward pass starts from the **'end'** node and move towards the **'start'** node

- Used to calculate latest occurrence times of all the activities

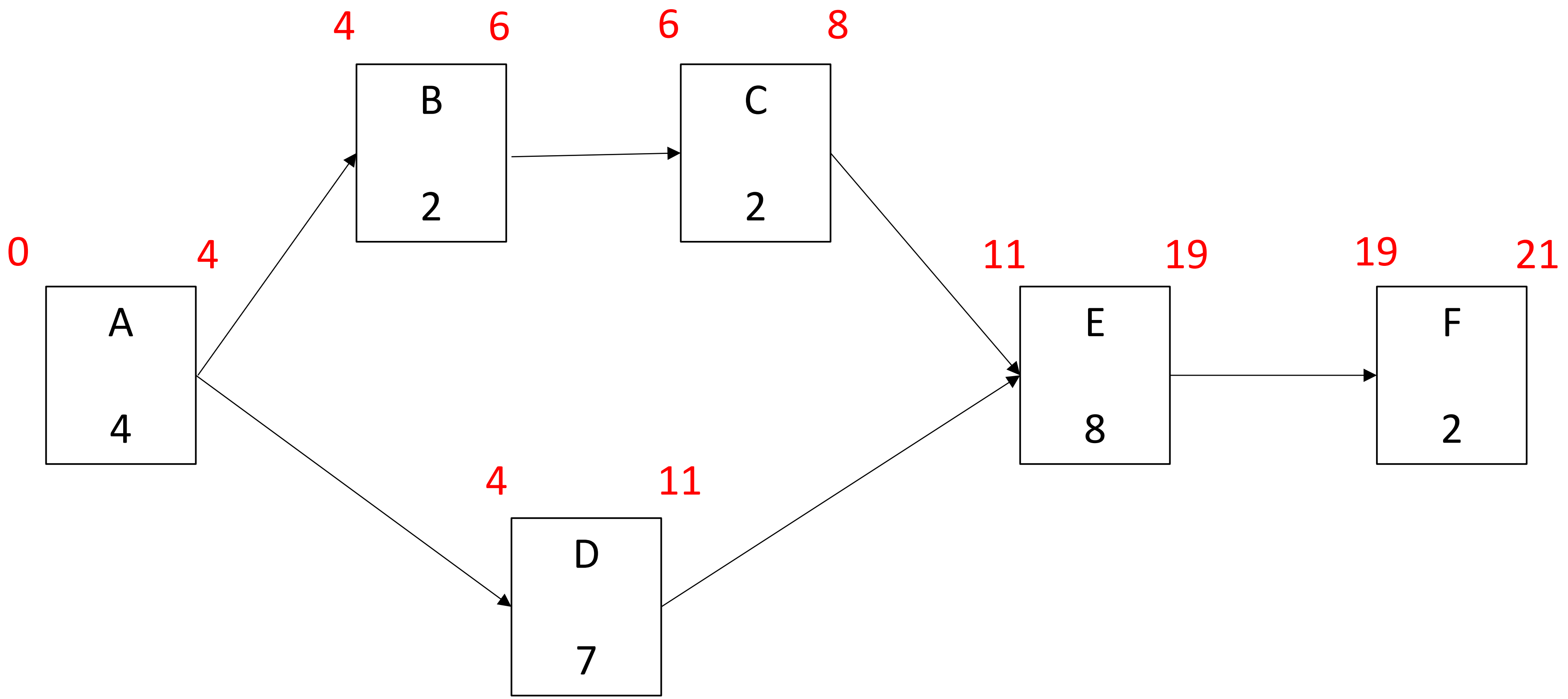
- We consider that for the last node LFT = EFT.

- The late occurrence time is given by

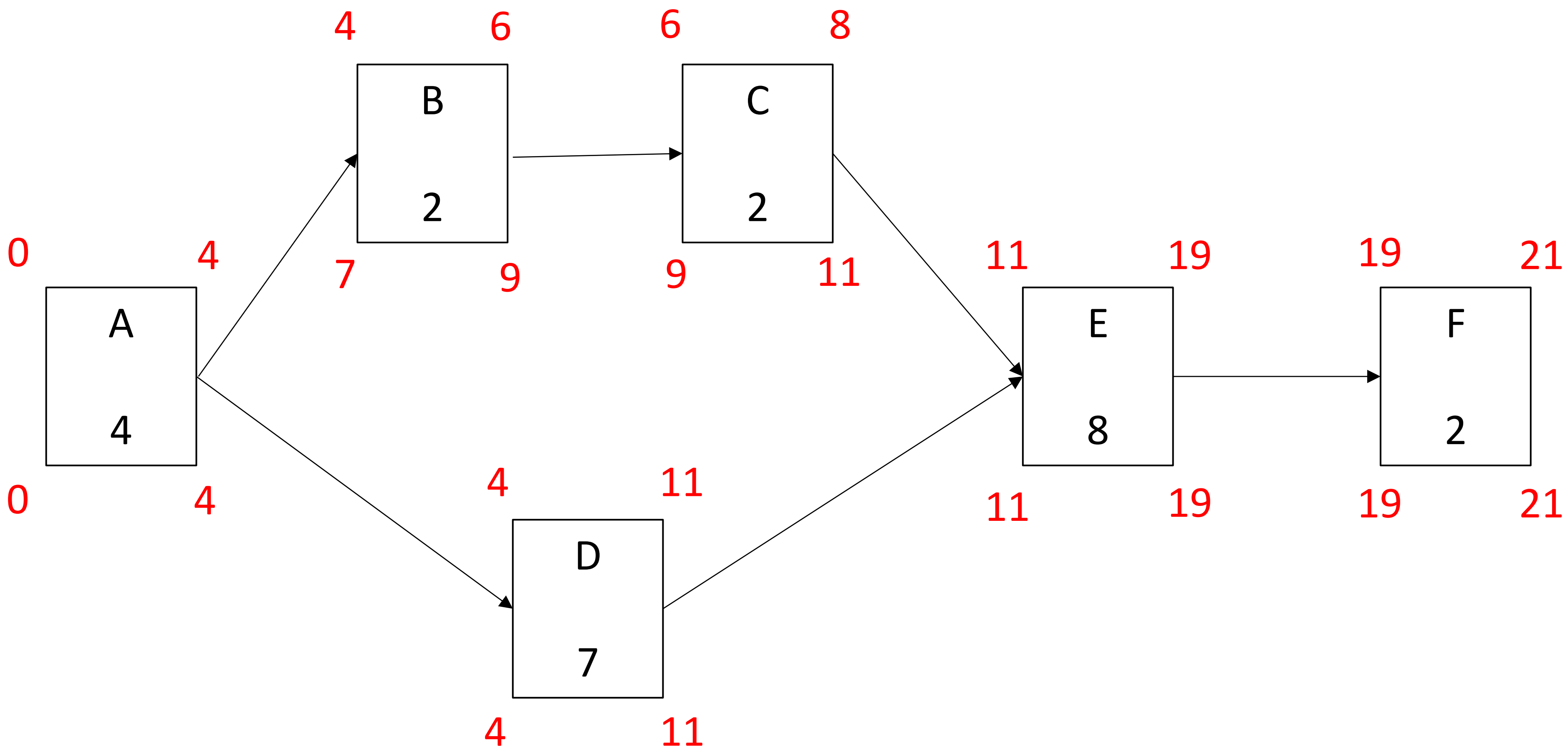
$$L_i = \text{Min}\ (L_j\ - \text{Dur})$$

where **Minimization is over all nodes j that succeed node i**

Exercise 4 – Carry out backward pass on this network



Solution 4 – Carry out backward pass on this network



Critical activities and Float

- The activities where there is no difference between the start times and finish times is called critical activities

- Non-critical activities – above condition is not met

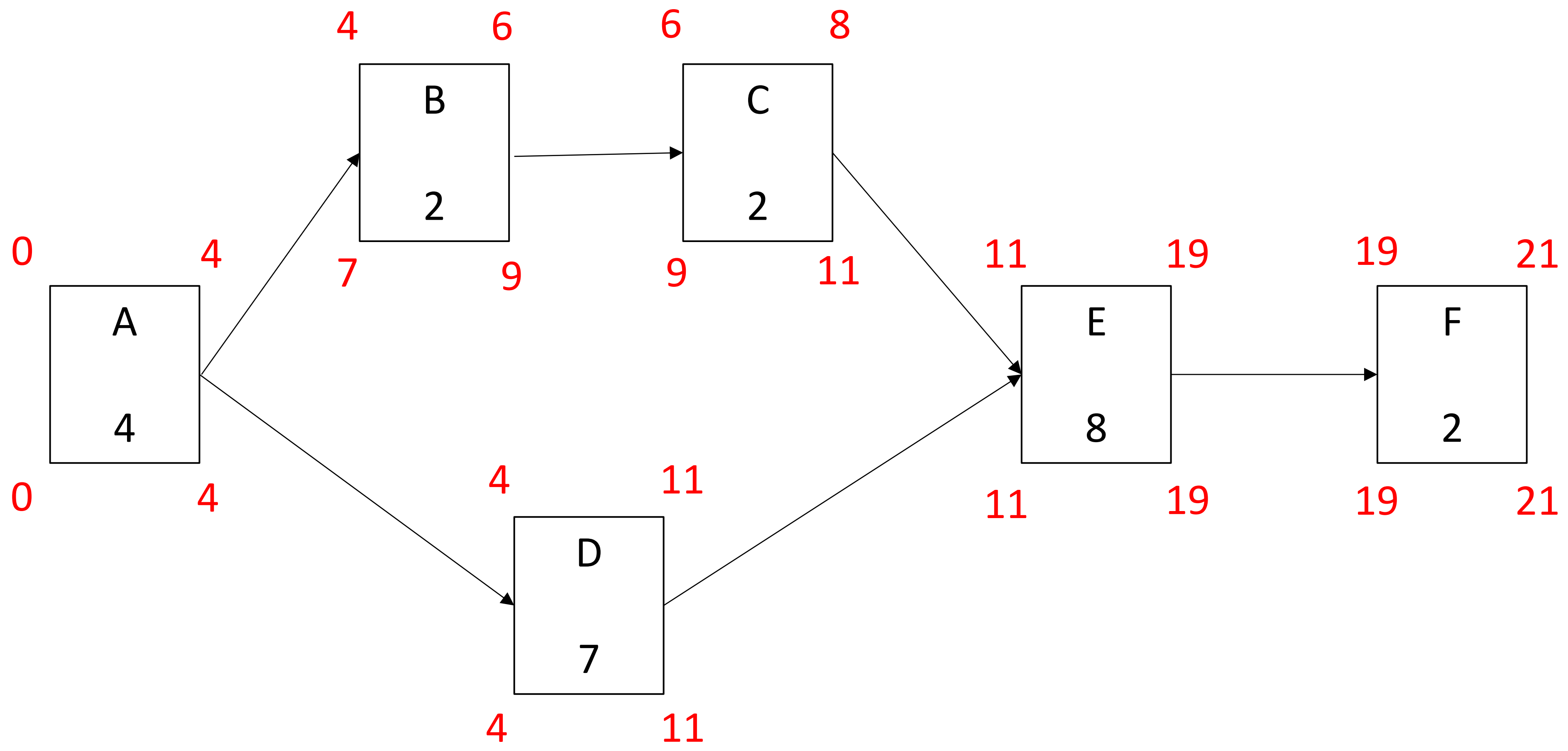
- Float or slack time –

$$\text{Total float} = LST_i - EST_i$$

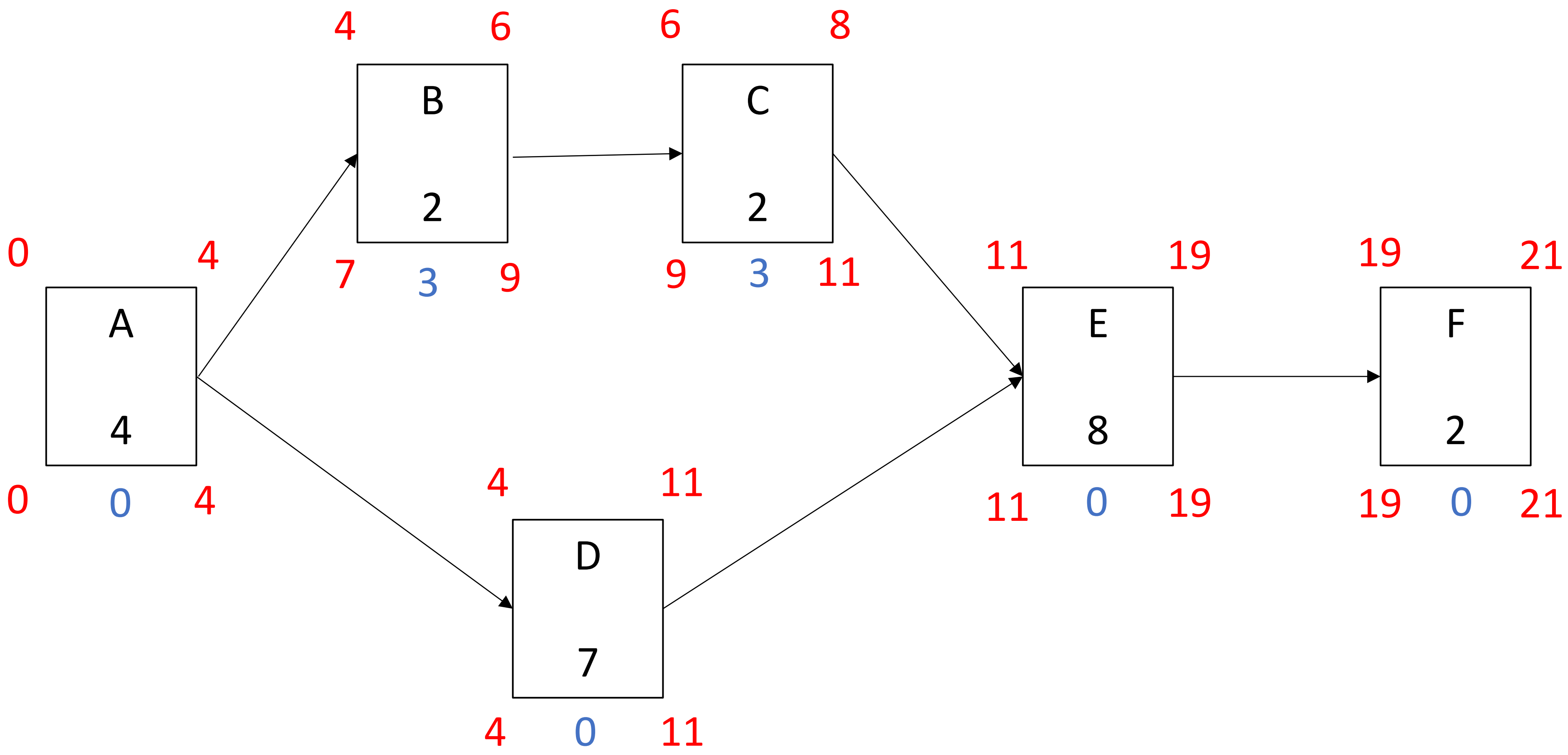
$$\text{or} \qquad \text{Total float} = LFT_i - EFT_i$$

- **Critical path** – set of activities connecting start and end node and having zero float or slack time.

Exercise 5 – Calculate Total floats for this network

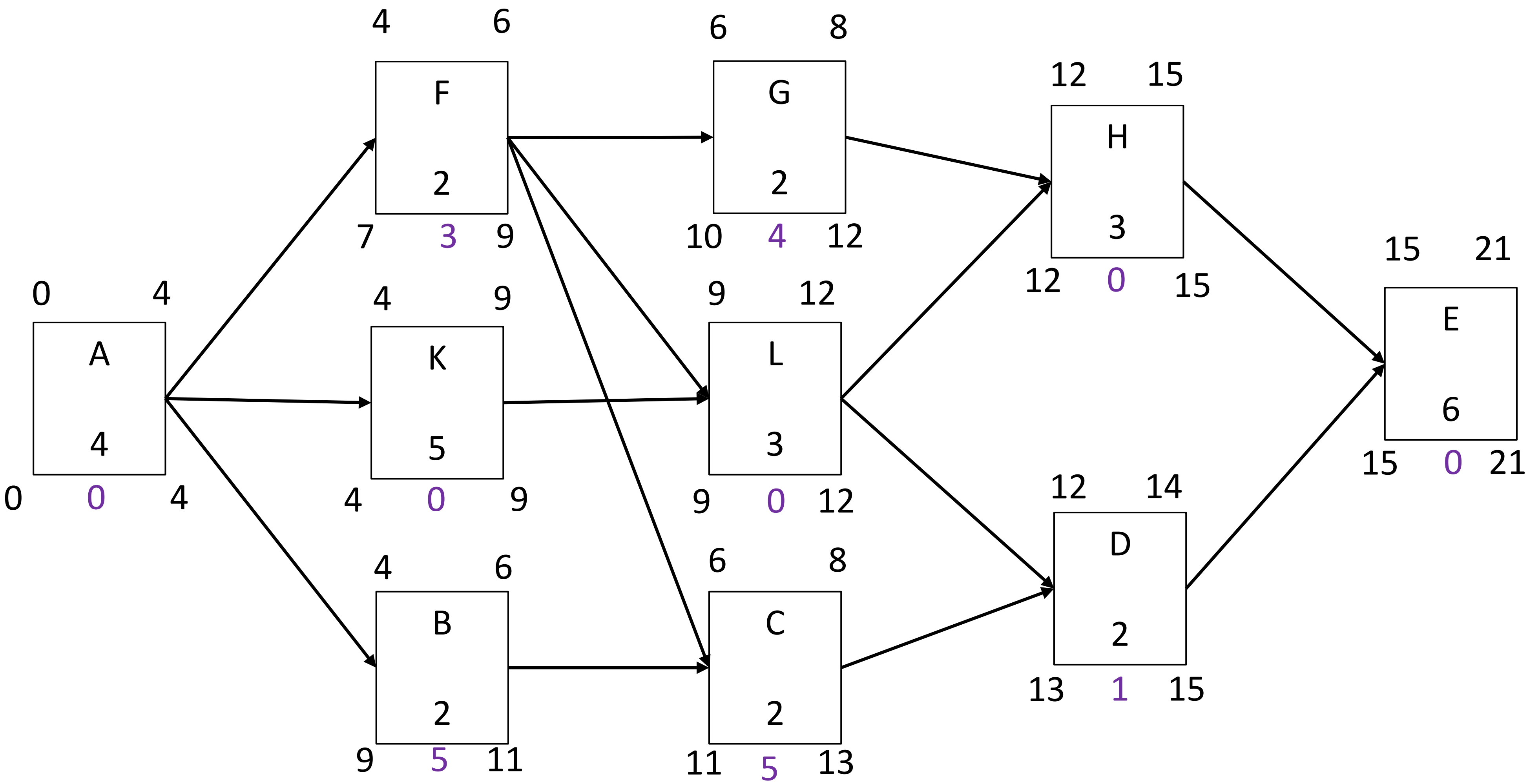


Solution 5 – Calculate Total floats for this network



What is the critical path?

Solution for Exercise 6



Different types of float

Types of Float or Slack time

- **Total float** – time an activity can be delayed without causing a delay in the completion of the entire project

- **Free float** – time by which start of an activity can be delayed without delaying the start of the following activity

- **Independent float –** time by which start of an activity maybe delayed without affecting the preceding or the following activity

- **Interference float** = Total float – free float

How can you calculate these Floats?

- **Total float** – time an activity can be delayed without causing a delay in the completion of the entire project
- **Free float** – time by which start of an activity can be delayed without delaying the start of the following activity

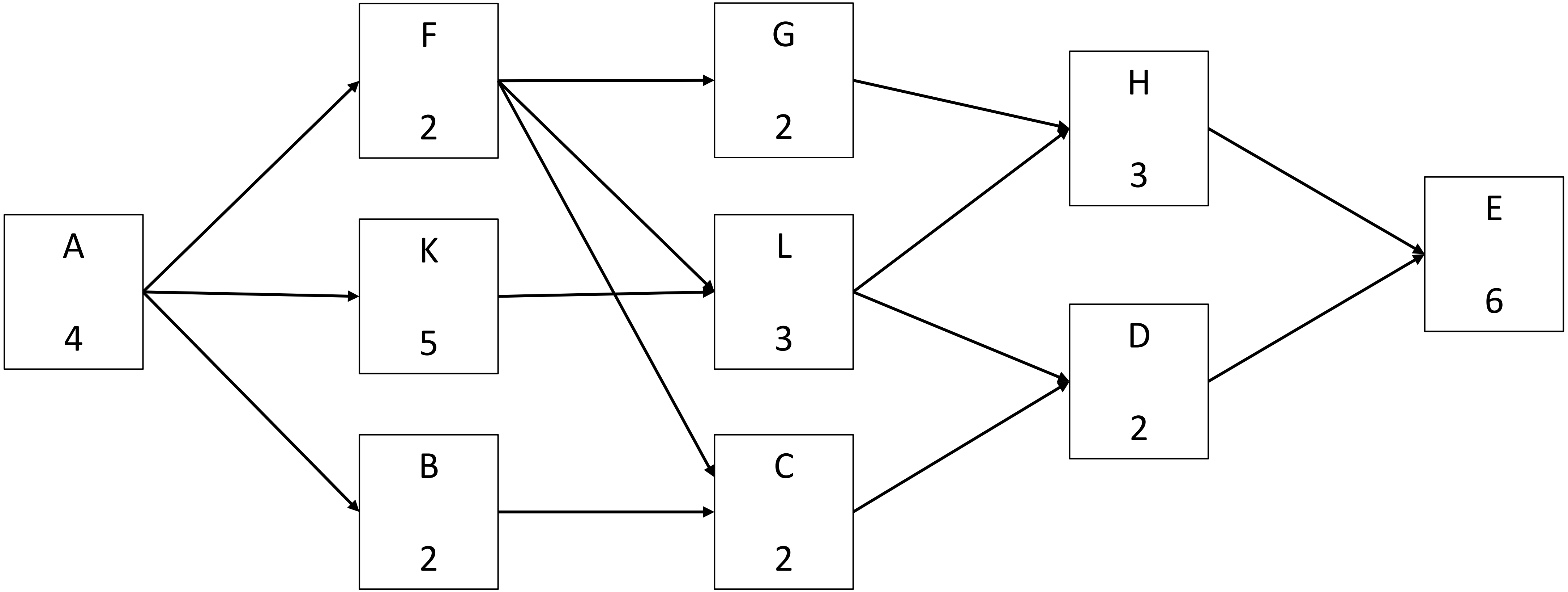
$$FF\,(i,j) = E_j - E_i - D(i,j)$$

- **Independent float –** time by which start of an activity maybe delayed without affecting the preceding or the following activity (including their floats)

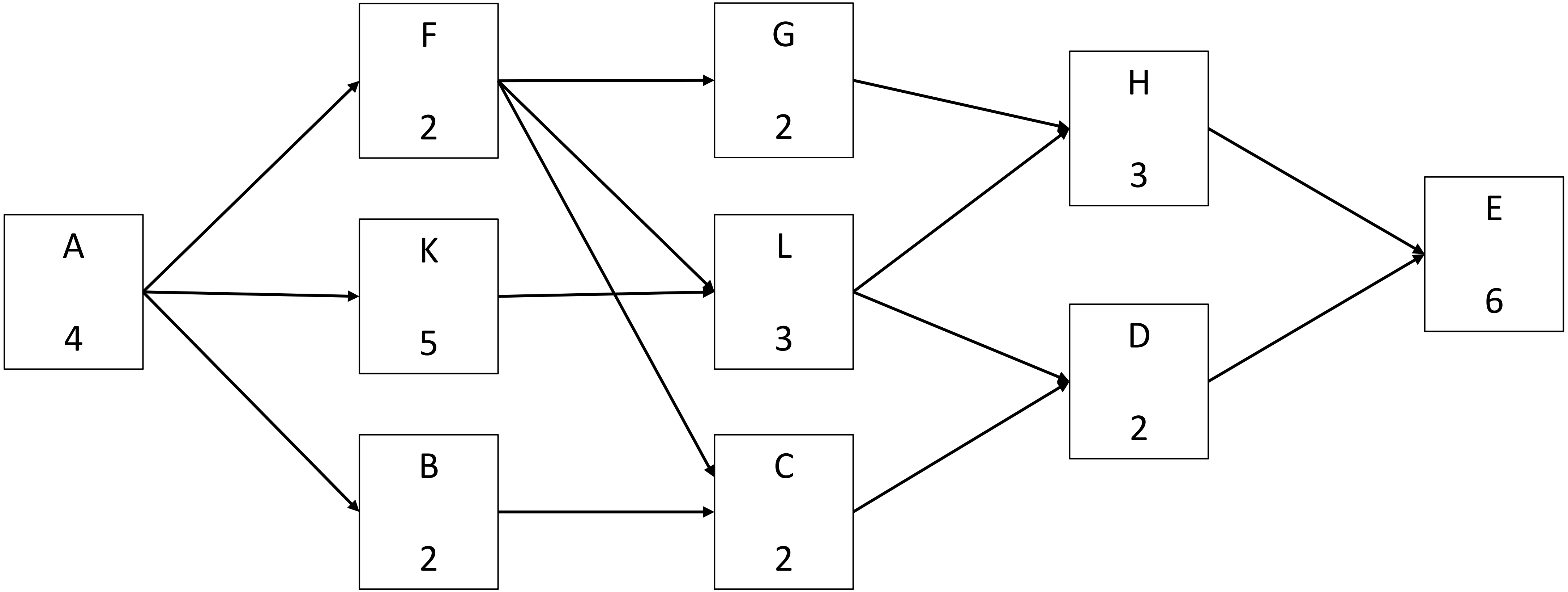
$$IF\,(i,j) = E_j - L_i - D(i,j)$$

- **Interference float** = Total float – free float

Exercise 6 – Compute total duration, floats, and critical path for this network



How can we update this schedule at Day 12?



THANK YOU